October-16

Day-13

SUBNETTING (PART-I)

- CIDR lets us be more flexible with IPv4 and networks

IPv4 Address Classes

Class	First octet (binary)	First octet range (decimal)
A	0xxxxxxx	0-127 (0.0.0.0 – 127.25
B	10xxxxxx	128-191 (128.0.0.0 – 191.255.255.2
C	110xxxxx	192-223 (192.0.0.0 – 223.255.255.25
D	1110xxxx	224-239 (224.0.0.0 – 239.255.25
E	1111xxxx	240-255 (240.0.0. 255.255.25

Only class A, B & C addresses can be assigned to a device as an IP address. as classes D & E have special purposes.
Class A, B & C have prefix length of /8, /16 & /24 respec

IPv4 address assignment.

How does a company get their own network address range to use?

↳ IP addresses are assigned to companies or organizations by non-profit American corporation called the 'IANA'.

- IANA (Internet Assigned Numbers Authority) assigns IPv4 address /networks to companies based on their size.

- For e.g very large company might receive class A or class B network, while small company might receive class C network.

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- However this led to many wasted IP addresses.
  How IP addresses gets wasted by this method?
- Company X needs IP addressing for 5000 end hosts.
- A class C network doesn't provide enough addresses, so a class B network must be used
- This will result in about 60,000 addresses being wasted.

CIDR (Classless Inter-Domain Routing)

- When the Internet was first created, the creators didn't predict that the Internet would be as large as it is today.
- This resulted in wasted IP address space like we learned in above eg.
- The total IPv4 address include over 4 billion addresses and that seemed like a huge no. of addresses when IPv4 was created, but now address space exhaustion is big problem as there's not enough addresses. One way to solve this problem CIDR.

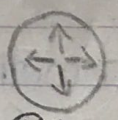
- The IETF (Internet Engineering Task Force) introduced CIDR in 1993 to replace 'classful' addressing system.

- With CIDR the requirement of class A = /8, class B = /16 & class C = /24 were removed.
- This also allowed larger networks to be split into smaller networks, allowing greater efficiency.

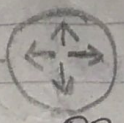
- These smaller networks are called 'subnetworks' or 'subnet'

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(Before concept of CIDR, yo tala ko network ma 254 vanda 252 vanda IP waste hunthyo)



203.0.113.0/24



R1 R2

11001011 . 00000000 . 01110001 . 00000000
203 . 0 . 113 . 0

11111111 . 11111111 . 11111111 . 00000000
255 . 255 . 255 . 0

No. of usable address = $2^8 - 2 = 254$
(network, broadcast)

But we only need 2, 1 for R1 & other for R2

- CIDR allows us to assign different prefix lengths, it doesn't have to be /24.

- Let's get some practice calculating the no. of hosts (usable address), with different prefix length.

a 203.0.113.0/25
b 203.0.113.0/26
c 203.0.113.0/27
d 203.0.113.0/28
e 203.0.113.0/29
f 203.0.113.0/30
g 203.0.113.0/31
h 203.0.113.0/32.

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a) 203.0.113.0/25

11001011 . 00000000 . 01110001 . 00000000
203 . 0 . 113 . 0 → part of network portion
11111111 . 11111111 . 11111111 . 10000000
255 . 255 . 255 . 128

Now, there are 7 bits in host portion of address. So the no. of usable address is $2^7 - 2 = 126$ usable address.

But we only need 2 addresses, 1 for R1 & other for R2. So, only 124 addresses gets wasted instead of 254 which is at least better.

b) 203.0.113.0/26.

Host - 11111111.11111111.11111111.11000000 = 255.255.255.192

No. of usable address = $2^6 - 2 = 62$ usable address

c) 203.0.113.0/27

Host: 11111111.11111111.11111111.11100000 = 255.255.255.224

No. of usable address = $2^5 - 2 = 30$

d) No. of usable address = $2^4 - 2 = 14$ (255.255.255.224).

e) No. of usable address = $2^3 - 2 = 6$ (255.255.255.240).

f) No. of usable address = $2^2 - 2 = 2$ (0 address will be waste if we use this in our earlier network).

It includes addresses from 203.0.113.0 - 203.0.113.3

It is the subnet of larger class C network

→ The remaining address in the 203.0.113.0/24 address block (203.0.113.4 - 203.0.113.255) are now available to be used in other network

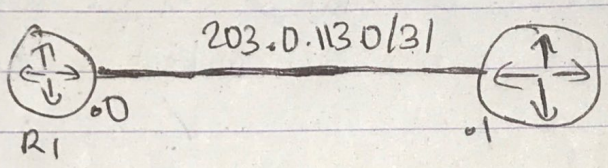
CIDR (/31)

↳ If we use a /31 prefix length, the mask is written as 255.255.255.254 in dotted decimal. There is now only 1 host bit, so that means 0 usable address.

$2^1 - 2 = 0$ usable address

So, you used to not be able to use /31 network prefixes because of this. HOWEVER, for a point to point connection like this, it's actually possible to use /31 mask.

(अनधिकार network)



The 203.0.113.0/31 consists of address 203.0.113.0 & 203.0.113.1. Normally this would be problem as, there would be no usable address after subtracting network & broadcast address.

But for point-point network like this a dedicated connection between two routers, there is actually no need for broadcast address or netwo. So, we can break rules & assign the only 2 address in this network to our two routers. Cisco में configure यदि we get warning but its valid. /31 is more efficient than /30

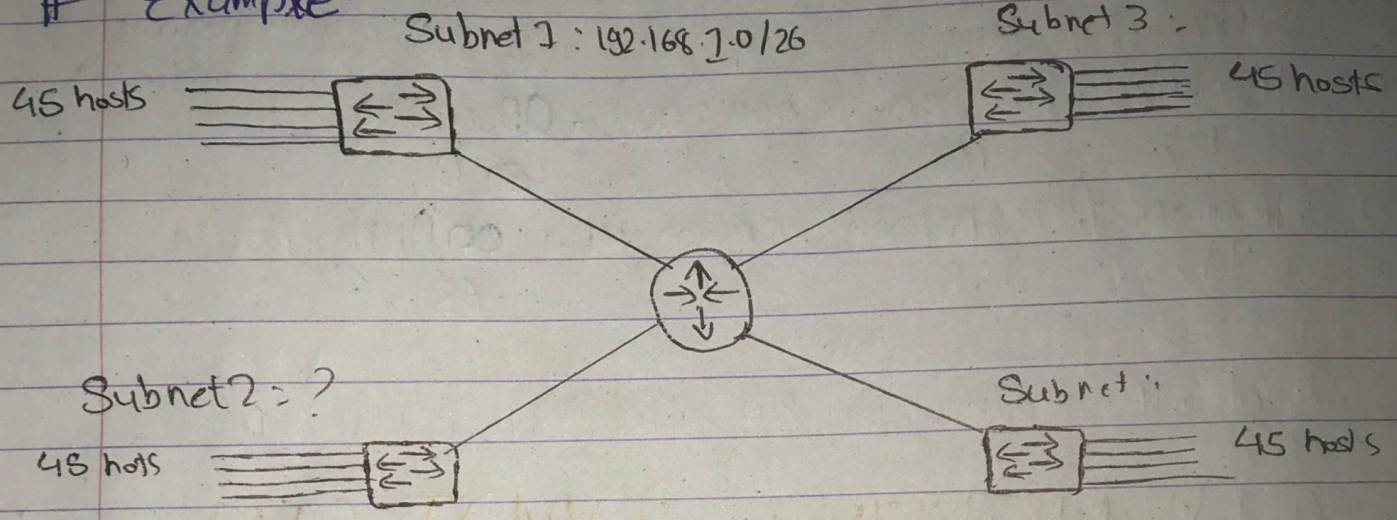
CIDR (/32)

Host: 255.255.255.255

We'll probably never use /32 to configure interface, However, when we want static route to just one host we can use /32 mask to specify that host

Class B, class A network को पनि subnetting हुन्छ with prefix length like /7, /19, /12 etc.

Example:



Divide the 192.168.1.0/24 network into four subnets that can accomodate the no. of hosts required.

↳ We need 4 equal size subnets with enough room for at least 45 hosts in each subnet.

We can't always make subnets have exactly the no. of addresses as we want. There might be some unused address space. It's fine coz its always good to have some room for growth.

Let's calculate subnet now,

~~$2^2 - 2$~~ i.e $/27 = 2^5 - 2 = 30$ usuable address

$/26 = 2^6 - 2 = 62$ usable address (suitable for us)

QUIZ

Q.1)a) In above fig, what are remaining subnets? (Hint. The next address to broadcast of subnet 1 is the network address for subnet 2 & so on)

Subnet 1 : 192.168.1.0 - 192.168.1.63

Subnet 2 : 192.168.1.64/26 192.168.1.64 - 192.168.1.127

Soln

Subnet 3 : 192.168.1.128/26 192.168.1.128-192.168.1.191

Subnet 4 : 192.168.1.192/26 192.168.1.192-192.168.1.255

For

Find network & broadcast of S1

/26 on network

11000000.10101000.00000001.00000000

192 . 168 . 1 . 0

11000000.10101000.00000001.00111111

192. 168 . 1 . 63

So, network address for S1 subnet 2 is 192.168.1.63